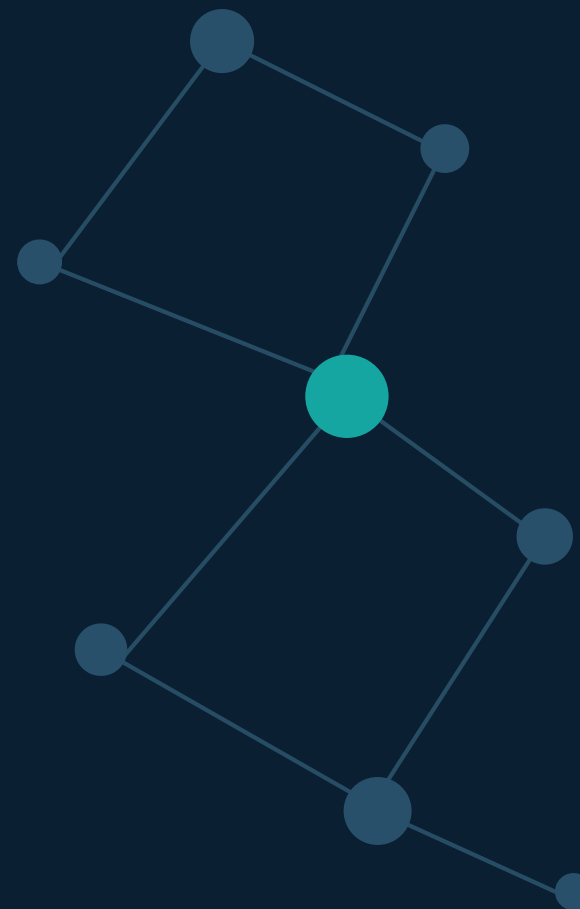


UNDERSTANDING THE DATA

Kidney Transplant Waitlist Dataset

A simple but technical walkthrough of what every row, feature, outcome, and data-quality issue means.

**494,862 waitlist registrations • 414,187 patients • 2015–
Jun 2026**



1. What exactly is this dataset?

A kidney-only analytic cohort built from OPTN waitlist data.

494,862

Waitlist registrations

One row = one registration

414,187

Unique patients

Some patients appear more than once

38

Columns

Clinical, timing, outcome, and ID fields

256

Listing centers

Centers vary greatly in size and outcomes

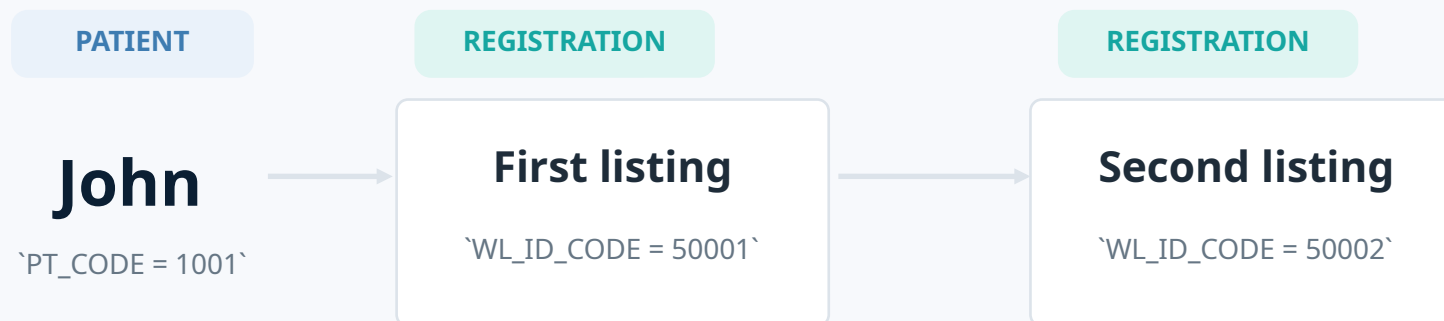
Cohort definition

Kidney-only waitlist registrations with Listing Date (`INIT\_DATE`) from January 1, 2015 onward. The latest listing date in the analytic file is June 30, 2026.

Simple meaning: This is a focused project cohort — not the entire OPTN database.

2. One row is NOT necessarily one person

The dataset is organized by waitlist registration.



Why this matters later

If the same patient appears in both training and testing, the model can indirectly “see” the same person twice.

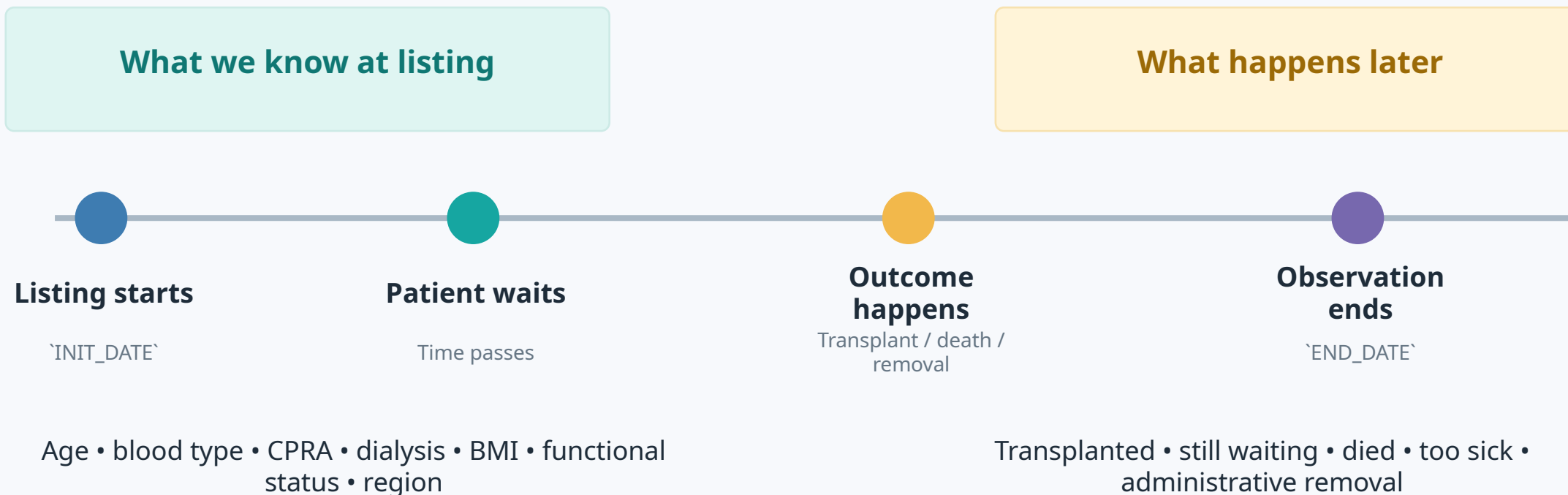
That is patient leakage.

Most patients appear once, but repeated registrations are common: 52,358 patients appear twice and 12,220 appear three or more times.

Simple meaning: Train/test splitting should be grouped by Patient ID (`'PT_CODE'`), not randomly by row.

3. Think of each row as a patient timeline

The project tries to predict the future using only information available near listing.



Simple meaning: Good prediction means using information from the LEFT side to predict what happens on the RIGHT — without leaking future information.

4. The 10 core listing-time features

These are the main variables considered safe for prediction at the time of listing.

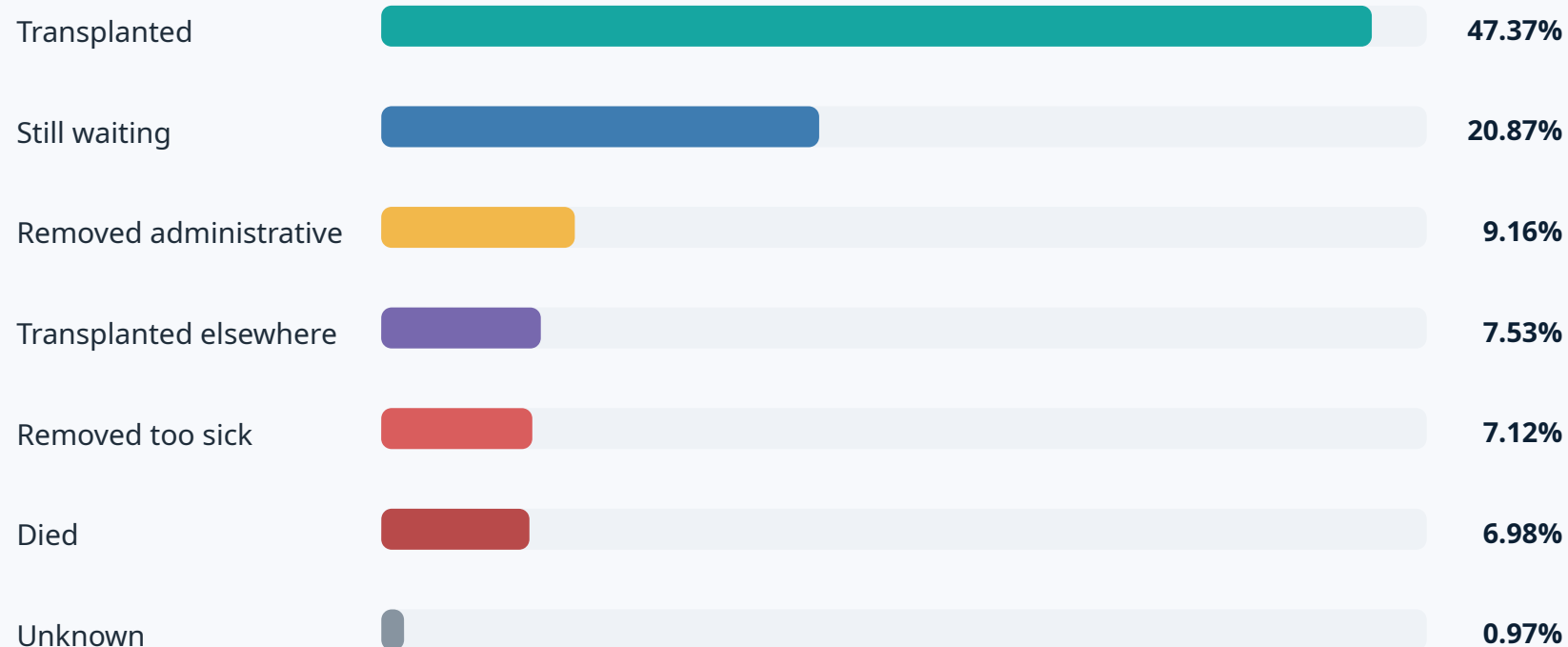
 Age at Listing `INIT_AGE`	 Sex `GENDER`	 Race / Ethnicity `ETHCAT`	 Blood Type `ABO`	 Initial CPRA `INIT_CPRA`
 On Dialysis `ON_DIALYSIS`	 Body Mass Index `BMI_TCR`	 Functional Status `FUNC_STAT_TCR`	 Initial Medical Status `INIT_STAT`	 OPTN Region `REGION`

Technical idea These become model inputs (X). The outcome we want to predict becomes the target (y).

Simple meaning: The key question is not “what columns exist?” — it is “what columns would truly be known at listing?”

5. What happened to patients?

Outcome (`outcome`) has seven possible categories.



Primary target
Adverse Event
`event\_adverse`

1 = died OR removed too sick

69,743
registrations
14.09%

Simple meaning: `event\_adverse` = 0 does not always mean “safe” — some patients are simply still waiting and have not had enough follow-up yet.

6. Censoring: the biggest time problem

A patient who is still waiting has an unfinished story.

Patient A

Listed 2015 → observed for years → no adverse event

Patient B

Listed May 2026 → observed for weeks → still waiting

Both currently have ``event_adverse = 0`` — but they are not equally strong negatives.



Simple meaning: Recent years look “safer” mainly because those patients have had less time for outcomes to occur. That is why survival analysis matters.

7. Initial CPRA: important clinically, tricky statistically

CPRA is a 0–100 measure of how difficult donor matching may be because of antibodies.

CPRA $\approx$ 0 Compatible with more potential donors

CPRA $\approx$ 100 Compatible with fewer potential donors

The surprise was not the values — it was the missingness.



Simple meaning: Missing CPRA is strongly tied to time. Do not blindly median-impute it as if the missing values were random.

8. Dialysis explains a major wait-time puzzle

Allocation priority time can be longer than the official registration time.

15.51%

Adverse rate: on dialysis

``ON_DIALYSIS = Y``

10.28%

Adverse rate: not on dialysis

``ON_DIALYSIS = N``

318,009

Started dialysis before listing

Many had already spent months/years on dialysis

Why can ``DAYSWAIT_ALLOC`` be larger than ``DAYSWAIT_CHRON``?

Official waitlist time

+

Eligible dialysis time before listing

≈

Allocation priority time

295,750 registrations matched exactly; 296,169 matched within 30 days.

Simple meaning: For most patients, the “extra” allocation credit is explained by dialysis time that began before official listing.

9. BMI exposed a real data-quality problem

The median looks normal — the extremes do not.

28.53

Median BMI

Looks plausible

263

BMI > 100

Clearly suspicious

124

BMI > 1,000

Impossible values

≈430k

Maximum BMI

Definitely a data error

EDA lesson A mean or standard deviation can become meaningless when a small number of impossible values are huge. That is why we inspected distributions and ranges instead of trusting summary statistics blindly.

Important: low-age patients are real pediatric candidates, but the extreme BMI values were mostly found in adults — so they are not explained by children.

Simple meaning: EDA finds the problem. Preprocessing decides how to handle it — likely by flagging or replacing clearly impossible BMI values with missing.

10. Which listing-time features showed the strongest patterns?

These are raw associations — useful signals, not proof of causation.

Age at Listing (`INIT\_AGE`)



Functional Status (`FUNC\_STAT\_TCR`)



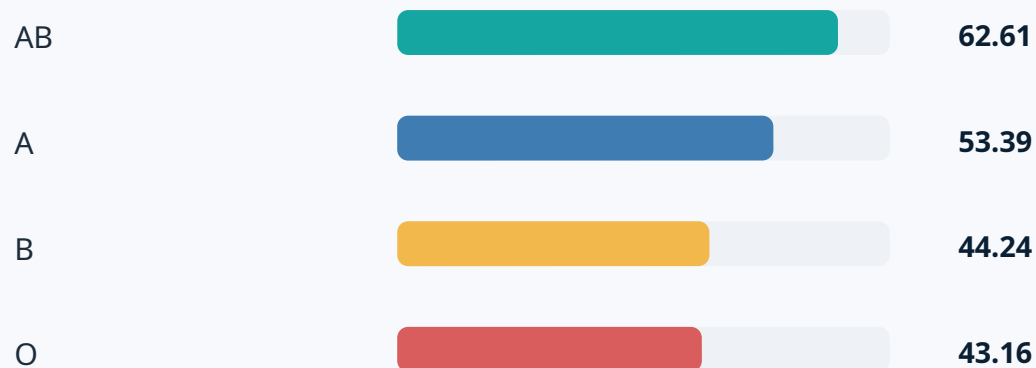
Dialysis statusOn dialysis: 15.51% adverse • Not on dialysis: 10.28%

Simple meaning: Older age, poorer functional status, and being on dialysis were among the clearest adverse-outcome signals in the EDA.

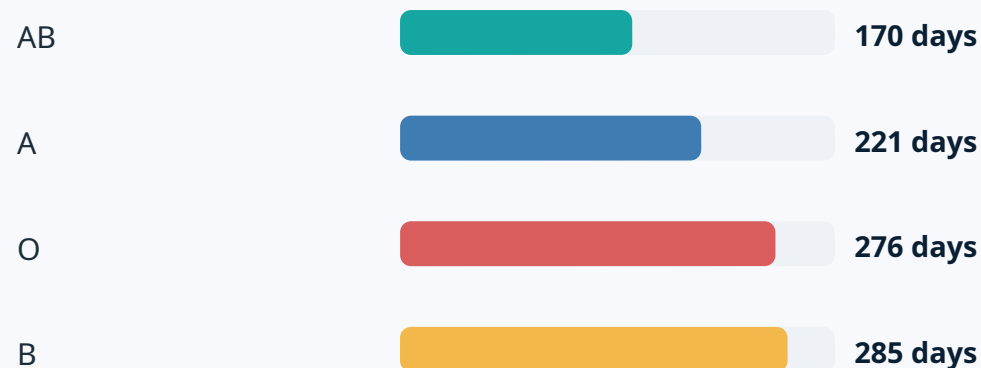
11. Blood type matters most for the transplant pathway

Observed transplant rates and waiting times differ clearly by Blood Type (`ABO`).

Observed transplant rate



Median wait among transplanted



Important nuance: rare subtypes such as A1B and A2B had very small sample sizes, so their percentages are less stable.

Simple meaning: In this dataset, B and O had lower observed transplant rates and longer waits than A and AB.

12. Geography and transplant center both matter

System-level variation is one of the strongest themes in the dataset.

Observed transplant rate by OPTN region



Region gap

55.62% → 42.47%

More than 13 percentage points

Center-level variation (centers with ≥ 500 registrations)

Adverse rate: 2.72% → 34.55%
Transplant rate: 16.04% → 81.60%

Simple meaning: These are raw differences. They show heterogeneity worth studying — not proof that one region or center is “better” or “worse.”

13. Missing data is not one single problem

We found three very different kinds of missingness.

1. Real missing data

Example: Body Mass Index ('BMI_TCR') missing because the measurement was unavailable.

Usually needs imputation or a missing-value strategy.

2. Structural missingness

Example: Dialysis Start Date missing because the patient is not on dialysis.

Missing is meaningful and often should stay meaningful.

3. Time / system missingness

Example: Initial CPRA becomes ~80% missing after 2022.

Requires investigation; ordinary imputation can hide a system change.

Simple meaning: A missing value only makes sense after we understand why that field can be missing.

14. The dangerous columns: future information

A model can look amazing and still be useless if we accidentally give it the answer.

SAFE near listing

- Age at Listing (`INIT\_AGE`)
- Blood Type (`ABO`)
- Initial CPRA (`INIT\_CPRA`)
- Dialysis Status (`ON\_DIALYSIS`)
- BMI, Functional Status, Region

LEAKY / FUTURE

- End Date (`END\_DATE`)
- Transplant Date (`TX\_DATE`)
- End CPRA (`END\_CPRA`)
- Derived Time to Outcome (`days\_to\_event`)
- Outcome / removal / donor fields

Target leakage = giving the model information that would only be known after the prediction moment.

Simple meaning: Predicting from listing means pretending we are standing at `INIT\_DATE`. Anything learned later must stay out of the feature set.

15. Internal consistency checks mostly passed

The derived timing fields agree well with the source data, with a few clear exceptions.

494,798

Exact matches

``DAYSWAIT_CHRON` = `days_to_event``

5

Numeric mismatches

Both present but different

59

Impossible date records

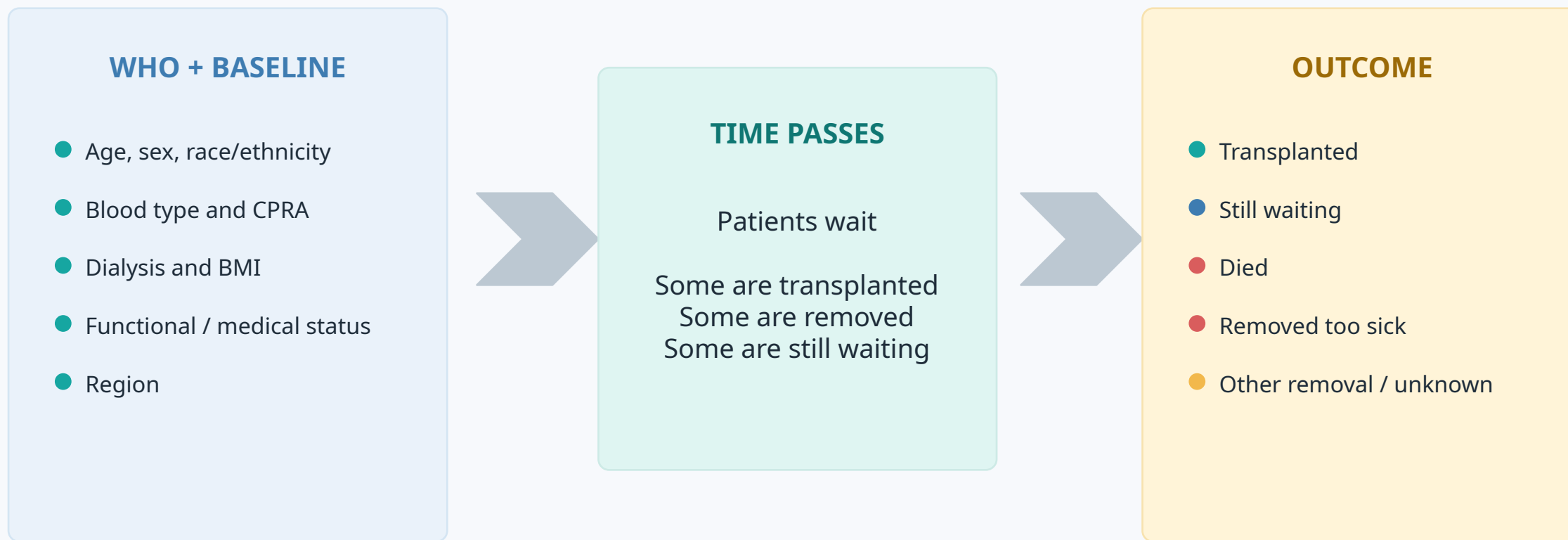
``END_DATE` < `INIT_DATE``

What this tells us The pipeline's Derived Time to Outcome (``days_to_event``) is very consistent with the source wait-time field. The 59 negative-duration registrations are genuine data errors and should not be used in survival-duration modeling unless corrected.

Simple meaning: EDA is not only “finding patterns” — it also validates whether our derived variables make sense.

16. The whole data story in one picture

This is the mental model to carry into preprocessing and modeling.



Simple meaning: Our job: learn from the left side, respect the time in the middle, and predict the right side without accidentally using future information.

17. Eight things to remember about the data

If you understand these, you understand the foundation of the project.

1

One row = one waitlist registration, not necessarily one person.

2

494,862 registrations represent 414,187 unique patients.

3

The main adverse target is death or removal because too sick.

4

Only 14.09% are adverse, so class imbalance matters.

5

20.87% are still waiting, so censoring matters.

6

Initial CPRA becomes ~80% missing after 2022.

7

BMI contains clearly impossible values that require cleaning.

8

Future / post-outcome columns must stay out of listing-time models.

Simple meaning: Next step: turn this understanding into a leakage-safe, well-documented preprocessing pipeline and model-ready dataset.

THE FOUNDATION IS NOW CLEAR

Understand the data first. Then trust the model.

The dataset is powerful, but only if we respect repeated patients, censoring, missingness, data-quality problems, and target leakage.

```
graph TD; A[Next] --> B[Preprocessing]; B --> C[Baseline models]; C --> D[Survival analysis]; D --> E[Fairness + explainability];
```

Next
Preprocessing
↓
Baseline models
↓
Survival analysis
↓
**Fairness +
explainability**